

Презентация по лабораторной работе №5

Моделирование сетей передачи данных

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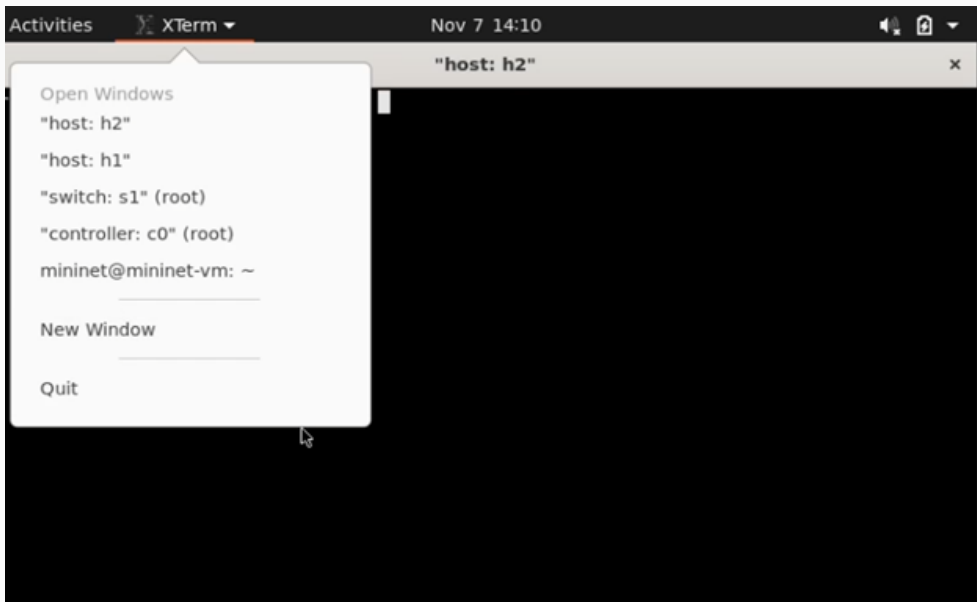
- Получить навыки проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных. Эти параметры влияют на производительность протоколов и сетей.

Выполнение лабораторной работы

```
mininet@mininet-vm:~$ xauth list $DISPLAY
mininet-vm/unix:11  MIT-MAGIC-COOKIE-1  e890b343e47c7b39dccf2f85690e8b2d
mininet@mininet-vm:~$ sudo -i
root@mininet-vm:~# xauth add mininet-vm/unix:11  MIT-MAGIC-COOKIE-1  e890b343e47c7b39dccf2f85690e8b2d
root@mininet-vm:~# xauth list
mininet-vm/unix:10  MIT-MAGIC-COOKIE-1  798c45b3820644e0b9cfb8c4cf8a6759
mininet-vm/unix:12  MIT-MAGIC-COOKIE-1  6a2c7f8e8b69ea601b845ba4952f5d02
mininet-vm/unix:11  MIT-MAGIC-COOKIE-1  e890b343e47c7b39dccf2f85690e8b2d
root@mininet-vm:~# logout
```

Рис. 1: Исправление прав запуска X-соединения в виртуальной машине mininet

Запуск лабораторной топологии



```
Activities XTerm Nov 7 14:10
"host: h2"
root@mininet-vm:/home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 0a:08:b4:61:82:d7 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#
```

```
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 6
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.41 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.159 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.048 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.048 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.035 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5097ms
rtt min/avg/max/mdev = 0.035/0.457/2.409/0.873 ms
root@mininet-vm:/home/mininet#
```

Рис. 4: Проверка подключения между хостами h1 и h2


```
rtt min/avg/max/mdev = 0.035/0.457/2.409/0.873 ms  
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 1  
0%  
root@mininet-vm:/home/mininet#
```

Рис. 5: Добавление 10% потерь пакетов на хосте h1

```
Activities  XTerm  Nov 7 14:15  "host: h1" x
64 bytes from 10.0.0.2: icmp_seq=78 ttl=64 time=0.056 ms
64 bytes from 10.0.0.2: icmp_seq=79 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=80 ttl=64 time=0.045 ms
64 bytes from 10.0.0.2: icmp_seq=81 ttl=64 time=0.046 ms
64 bytes from 10.0.0.2: icmp_seq=82 ttl=64 time=0.045 ms
64 bytes from 10.0.0.2: icmp_seq=83 ttl=64 time=0.045 ms
64 bytes from 10.0.0.2: icmp_seq=84 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=85 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=86 ttl=64 time=0.045 ms
64 bytes from 10.0.0.2: icmp_seq=87 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=88 ttl=64 time=0.045 ms
64 bytes from 10.0.0.2: icmp_seq=89 ttl=64 time=0.054 ms
64 bytes from 10.0.0.2: icmp_seq=90 ttl=64 time=0.031 ms
64 bytes from 10.0.0.2: icmp_seq=91 ttl=64 time=0.046 ms
64 bytes from 10.0.0.2: icmp_seq=92 ttl=64 time=0.085 ms
64 bytes from 10.0.0.2: icmp_seq=93 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=94 ttl=64 time=0.057 ms
64 bytes from 10.0.0.2: icmp_seq=95 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=96 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=97 ttl=64 time=0.050 ms
64 bytes from 10.0.0.2: icmp_seq=98 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=99 ttl=64 time=0.053 ms
64 bytes from 10.0.0.2: icmp_seq=100 ttl=64 time=0.048 ms

--- 10.0.0.2 ping statistics ---
100 packets transmitted, 86 received, 14% packet loss, time 101370ms
rtt min/avg/max/mdev = 0.027/0.057/0.471/0.053 ms
root@mininet-vm: /home/mininet#
```

```
rtt min/avg/max/mdev = 0.034/0.180/0.851/0.300 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h2-eth0 root netem loss 1
0%
root@mininet-vm:/home/mininet# █
```

Рис. 7: Добавление 10% потерь пакетов на хосте h2

```
"host: h1" x
64 bytes from 10.0.0.2: icmp_seq=75 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=76 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=77 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=79 ttl=64 time=0.041 ms
64 bytes from 10.0.0.2: icmp_seq=80 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=81 ttl=64 time=0.057 ms
64 bytes from 10.0.0.2: icmp_seq=82 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=83 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=84 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=85 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=86 ttl=64 time=0.084 ms
64 bytes from 10.0.0.2: icmp_seq=88 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=89 ttl=64 time=0.046 ms
64 bytes from 10.0.0.2: icmp_seq=90 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=91 ttl=64 time=0.040 ms
64 bytes from 10.0.0.2: icmp_seq=92 ttl=64 time=0.075 ms
64 bytes from 10.0.0.2: icmp_seq=94 ttl=64 time=0.034 ms
64 bytes from 10.0.0.2: icmp_seq=95 ttl=64 time=0.045 ms
64 bytes from 10.0.0.2: icmp_seq=96 ttl=64 time=0.048 ms
64 bytes from 10.0.0.2: icmp_seq=97 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=98 ttl=64 time=0.057 ms
64 bytes from 10.0.0.2: icmp_seq=99 ttl=64 time=0.037 ms
64 bytes from 10.0.0.2: icmp_seq=100 ttl=64 time=0.051 ms

--- 10.0.0.2 ping statistics ---
100 packets transmitted, 82 received, 18% packet loss, time 101353ms
rtt min/avg/max/mdev = 0.034/0.062/0.705/0.077 ms
root@mininet-vm:/home/mininet#
```

```
rtt min/avg/max/mdev = 0.034/0.062/0.705/0.077 ms  
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem  
root@mininet-vm:/home/mininet#
```

Рис. 9: Восстановление конфигурации по умолчанию для хоста h1 и хоста h2

```
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 100
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.733 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.913 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.137 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.040 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.042 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.051 ms
^C
--- 10.0.0.2 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6091ms
rtt min/avg/max/mdev = 0.040/0.280/0.913/0.348 ms
root@mininet-vm:/home/mininet#
```

Рис. 10: Проверка

```
rtt min/avg/max/mdev = 0.040/0.280/0.913/0.348 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 5
0% 50%
root@mininet-vm:/home/mininet# █
```

Рис. 11: Добавление на узле h1 коэффициента потери пакетов 50%

```
Activities  XTerm  Nov 7 14:22  "host: h1" x
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.068 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.054 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.045 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.038 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.104 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.038 ms
64 bytes from 10.0.0.2: icmp_seq=21 ttl=64 time=0.054 ms
64 bytes from 10.0.0.2: icmp_seq=24 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=25 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=26 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=27 ttl=64 time=0.042 ms
64 bytes from 10.0.0.2: icmp_seq=28 ttl=64 time=0.043 ms
64 bytes from 10.0.0.2: icmp_seq=31 ttl=64 time=0.045 ms
64 bytes from 10.0.0.2: icmp_seq=34 ttl=64 time=0.043 ms
64 bytes from 10.0.0.2: icmp_seq=35 ttl=64 time=0.043 ms
64 bytes from 10.0.0.2: icmp_seq=36 ttl=64 time=0.043 ms
64 bytes from 10.0.0.2: icmp_seq=37 ttl=64 time=0.041 ms
64 bytes from 10.0.0.2: icmp_seq=45 ttl=64 time=0.045 ms
64 bytes from 10.0.0.2: icmp_seq=48 ttl=64 time=0.042 ms

--- 10.0.0.2 ping statistics ---
50 packets transmitted, 26 received, 48% packet loss, time 50164ms
rtt min/avg/max/mdev = 0.038/0.080/0.540/0.100 ms
root@mininet-vm: /home/mininet#
```



```
rtt min/avg/max/mdev = 0.038/0.080/0.540/0.100 ms
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# sudo tc qdisc
```

Рис. 13: Восстановление конфигурации интерфейса по умолчанию

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem corrup  
t 0.01%  
root@mininet-vm:/home/mininet#
```

Рис. 14: Добавление на узле h1 0.01% повреждения пакетов

```

root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 55642 connected to 10.0.0.2 port 5201
[ ID] Interval            Transfer          Bitrate          Retr   Cwnd
[ 7]  0.00-1.00    sec  4.82 GBytes    41.4 Gbits/sec    15     701 KBytes
[ 7]  1.00-2.00    sec  4.38 GBytes    37.6 Gbits/sec     8     634 KBytes
[ 7]  2.00-3.00    sec  4.23 GBytes    36.4 Gbits/sec    15     703 KBytes
[ 7]  3.00-4.00    sec  4.44 GBytes    38.2 Gbits/sec    12     601 KBytes
[ 7]  4.00-5.00    sec  4.35 GBytes    37.4 Gbits/sec     7     611 KBytes
[ 7]  5.00-6.00    sec  4.67 GBytes    40.1 Gbits/sec     9     686 KBytes
[ 7]  6.00-7.00    sec  4.67 GBytes    40.1 Gbits/sec    11     638 KBytes
[ 7]  7.00-8.00    sec  4.90 GBytes    42.1 Gbits/sec    14     706 KBytes
[ 7]  8.00-9.00    sec  4.85 GBytes    41.7 Gbits/sec    12     380 KBytes
[ 7]  9.00-10.00   sec  4.04 GBytes    34.7 Gbits/sec     5     434 KBytes
- - - - -
[ ID] Interval            Transfer          Bitrate          Retr
[ 7]  0.00-10.00   sec  45.4 GBytes    39.0 Gbits/sec   108
[ 7]  0.00-10.00   sec  45.3 GBytes    38.9 Gbits/sec

iperf Done.
root@mininet-vm:/home/mininet#

```

```
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
```

Рис. 16: Восстановление конфигурации интерфейса по умолчанию

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay  
10ms reorder 25% 50%  
root@mininet-vm:/home/mininet#
```

Рис. 17: Добавление на узле h1 правила из лабораторной работы

```
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 20
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=10.2 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=10.5 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=10.2 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=10.1 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.045 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=10.5 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=10.2 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=10.5 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=11.6 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=10.2 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=10.1 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=10.1 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=10.5 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=10.3 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=10.1 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=10.2 ms
```

```
rtt min/avg/max/mdev = 0.044/9.333/11.631/3.116 ms  
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem  
root@mininet-vm:/home/mininet# █
```

Рис. 19: Восстановление конфигурации интерфейса по умолчанию

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem duplic  
ate 50%  
root@mininet-vm:/home/mininet#
```

Рис. 20: Добавление на узле h1 правила с дублированием 50% пакетов


```
Activities XTerm Nov 7 14:34
"host: h1"
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.052 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.041 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.044 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.055 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.059 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.039 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.040 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.053 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.054 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.055 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.067 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.067 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.055 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.042 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.046 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.056 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.056 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.065 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, +10 duplicates, 0% packet loss, time 19442 ms
rtt min/avg/max/mdev = 0.039/0.054/0.067/0.013 ms
```

```
rtt min/avg/max/mdev = 0.039/0.094/0.800/0.143 ms  
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem  
root@mininet-vm:/home/mininet# █
```

Рис. 22: Восстановление конфигурации интерфейса по умолчанию

```
0 upgraded, 0 newly installed, 0 to remove and 374 not upgraded.  
mininet@mininet-vm:~$ mkdir -p ~/work/lab netem ii/expname
```

Рис. 23: Создание каталога expname

```
mininet@mininet-vm:~$ mkdir -p ~/work/lab_netem_ii/simple-drop
mininet@mininet-vm:~$ cd ~/work/lab_netem_ii/simple-drop
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ touch lab_netem_ii.py
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ ls
lab_netem_ii.py
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$
```

Рис. 24: Создание каталога simple-drop и дальнейшее его открытие

Воспроизведение экспериментов

```
mininet@mininet-vm: ~/work/lab_netem_ji/simple-drop
GNU nano 4.8 lab_netem_ji.py Modified
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info
import time

def emptyNet():

    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller,waitConnected=True )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s1 = net.addSwitch( 's1' )
    info( '*** Creating links\n' )
    net.addLink( h1, s1 )
    net.addLink( h2, s1 )

    info( '*** Starting network\n' )
    net.start()

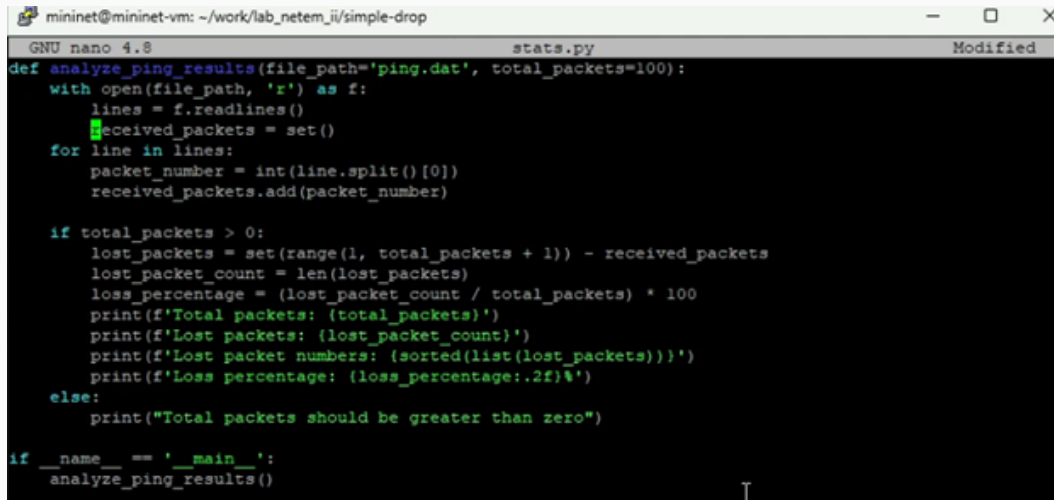
    info( '*** Set delay\n' )
    h1.cmdPrint( 'tc qdisc add dev h1-eth0 root netem loss 10%' )
    h2.cmdPrint( 'tc qdisc add dev h2-eth0 root netem loss 10%' )

    time.sleep(10) # Wait 10 seconds

    info( '*** Ping\n' )
    h1.cmdPrint( 'ping -c 100', h2.IP(), '| grep "time=" | awk \'{print $5, $7}\'' | sed -e \'s/time

    info( '*** Stopping network' )
    net.stop()

if __name__ == '__main__':
    setLogLevel( 'info' )
    emptyNet()
```

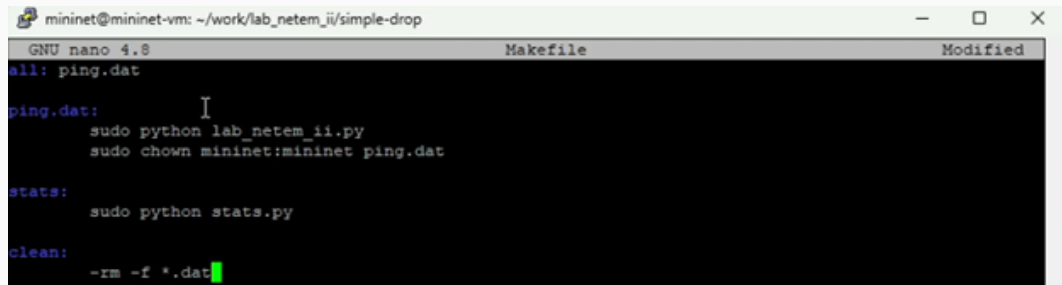


```
mininet@mininet-vm: ~/work/lab_netem_ii/simple-drop
GNU nano 4.8 stats.py Modified
def analyze_ping_results(file_path='ping.dat', total_packets=100):
    with open(file_path, 'r') as f:
        lines = f.readlines()
        received_packets = set()
    for line in lines:
        packet_number = int(line.split()[0])
        received_packets.add(packet_number)

    if total_packets > 0:
        lost_packets = set(range(1, total_packets + 1)) - received_packets
        lost_packet_count = len(lost_packets)
        loss_percentage = (lost_packet_count / total_packets) * 100
        print(f'Total packets: {total_packets}')
        print(f'Lost packets: {lost_packet_count}')
        print(f'Lost packet numbers: {sorted(list(lost_packets))}')
        print(f'Loss percentage: {loss_percentage:.2f}%')
    else:
        print("Total packets should be greater than zero")

if __name__ == '__main__':
    analyze_ping_results()
```

Рис. 26: Создание нового скрипта для вывода информации о потере пакетов



```
mininet@mininet-vm: ~/work/lab_netem_i/simple-drop
GNU nano 4.8 Makefile Modified
all: ping.dat

ping.dat:
    sudo python lab_netem_i.py
    sudo chown mininet:mininet ping.dat

stats:
    sudo python stats.py

clean:
    -rm -f *.dat
```

Рис. 27: Создание Makefile и помещение в него скрипта

Воспроизведение экспериментов

```
mininet@mininet-vm: ~/work/lab_netem_i1/simple-drop
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay
*** h1 : ('tc qdisc add dev h1-eth0 root netem loss 10%,')
*** h2 : ('tc qdisc add dev h2-eth0 root netem loss 10%,')
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "time=" |awk \'{print $5, $7}\'' | sed -e 's/time=//g\' -
e 's/icmp_seq=//g\' > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
mininet@mininet-vm:~/work/lab_netem_i1/simple-drop$ make stats
sudo python stats.py
  File "stats.py", line 4
    received_packets = set()
                        ^
TabError: inconsistent use of tabs and spaces in indentation
make: *** [Makefile:8: stats] Error 1
mininet@mininet-vm:~/work/lab_netem_i1/simple-drop$ nano stats.py
mininet@mininet-vm:~/work/lab_netem_i1/simple-drop$ make stats
sudo python stats.py
  File "stats.py", line 4
    received_packets = set()
                        ^
TabError: inconsistent use of tabs and spaces in indentation
make: *** [Makefile:8: stats] Error 1
mininet@mininet-vm:~/work/lab_netem_i1/simple-drop$ nano stats.py
mininet@mininet-vm:~/work/lab_netem_i1/simple-drop$ make stats
sudo python stats.py
Total packets: 100
Lost packets: 19
Lost packet numbers: [6, 8, 9, 16, 18, 23, 30, 35, 43, 50, 53, 66, 71, 87, 93, 95, 96, 97, 100]
Loss percentage: 19.00%
mininet@mininet-vm:~/work/lab_netem_i1/simple-drop$ ls
lab_netem_i1.py Makefile ping.dat stats.py
mininet@mininet-vm:~/work/lab_netem_i1/simple-drop$ make clean
```


Воспроизведение экспериментов

```
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ make
sudo python lab_netem_ii.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay
*** h1 : ('tc qdisc add dev h1-eth0 root netem loss 50% 50%,)
*** h2 : ('tc qdisc add dev h2-eth0 root netem loss 10%,)
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "time=" |awk \'(print $5, $7)\'| | sed -e \'s/time=//g\' -e \'s/icmp_seq=//g\' > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ make stats
sudo python stats.py
Total packets: 100
Lost packets: 60
Lost packet numbers: [1, 2, 3, 4, 7, 8, 9, 10, 11, 18, 21, 23, 24, 25, 30, 31, 32, 34, 35, 37, 39, 40, 41, 42, 43, 44, 54, 55, 58, 60, 61, 62, 63, 64, 68, 69, 71, 72, 77, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99]
Loss percentage: 60.00%
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ make clean
rm -f *.dat
```

Воспроизведение экспериментов

```
mininet@mininet-vm:~/work/lab_netem_i1/package-damage$ make
sudo python lab_netem_i1.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay
*** h1 : ('tc qdisc add dev h1-eth0 root netem corrupt 0.01%',)
*** h2 : ('tc qdisc add dev h2-eth0 root netem loss 10%',)
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "time=" |awk \'(print $5, $7)\' | sed -e \'s/time=//g\' -
e \'s/icmp_seq=//g\' > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
mininet@mininet-vm:~/work/lab_netem_i1/package-damage$ make stats
sudo python stats.py
Total packets: 100
Lost packets: 10
Lost packet numbers: [7, 17, 26, 32, 36, 37, 40, 41, 78, 92]
Loss percentage: 10.00%
mininet@mininet-vm:~/work/lab_netem_i1/package-damage$ make clean
```

Воспроизведение экспериментов

```
mininet@mininet-vm:~/work/lab_netem_ii/reordering-packages$ make
sudo python lab_netem_ii.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay
*** h1 : ('tc qdisc add dev h1-eth0 root netem delay 10ms reorder 25% 50%',)
*** h2 : ('tc qdisc add dev h2-eth0 root netem loss 10%',)
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "time=" |awk \'(print $5, $7)\' | sed -e \'s/time=//g\' -e \'s/icmp_seq=//g\' > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
mininet@mininet-vm:~/work/lab_netem_ii/reordering-packages$ make stats
sudo python stats.py
Total packets: 100
Lost packets: 9
Lost packet numbers: [19, 21, 22, 29, 38, 47, 59, 61, 67]
Loss percentage: 9.00%
mininet@mininet-vm:~/work/lab_netem_ii/reordering-packages$ make clean
```

Воспроизведение экспериментов

```
mininet@mininet-vm:~/work/lab_netem_ii/duplicating-packages$ make
sudo python lab_netem_ii.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay
*** h1 : ('tc qdisc add dev h1-eth0 root netem duplicate 50%',)
*** h2 : ('tc qdisc add dev h2-eth0 root netem ',)
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "time=" |awk \'(print $5, $7)\' | sed -e \'s/time=//g\' -
e \'s/icmp_seq=//g\' > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
mininet@mininet-vm:~/work/lab_netem_ii/duplicating-packages$ make stats
sudo python stats.py
Total packets: 100
Lost packets: 0
Lost packet numbers: []
Loss percentage: 0.00%
mininet@mininet-vm:~/work/lab_netem_ii/duplicating-packages$ make clean
rm -f *.dat
mininet@mininet-vm:~/work/lab_netem_ii/duplicating-packages$
```

Вывод

- В ходе выполнения лабораторной работы получили навыки проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных.

Список литературы. Библиография

[1] Mininet: <https://mininet.org/>